

Pre Submission Document

Group 26

Domain 10

Team members: Balachandra Vignesh Shunmugam, Rajesh Kumar, Sivasankari Ramasubramanian

1. Business Problem Understanding and Objective:

About the company: BuildMax Infrastructure does large construction projects and is facing frequent schedule delays, budget overruns and occurrence of safety incidents. Currently there is no proactive monitoring. Late projects and budget overspends cut into profits, lead to disputes and legal trouble, and damage the company's reputation.

The Business problem: Currently, BuildMax has no reliable way to identify risks related to delay in schedule or budget overrun or likely occurrence of safety incidents well ahead. Current data projects problems after they have occurred. We have to do a predictive analysis

2. Challenges and Constraints

Challenge / Constraint	What it means and how we handle it
Class imbalance	Only about 22% of projects in the data are delayed, need to handle this imbalance
Data quality issues	The data has duplicate rows, the same category written in different ways (one project type is spelled three ways), about 10% missing values in two columns, and at least one impossible cost value. We catch these while profiling the data and clean them up.
Non-informative columns	Some columns are just ID numbers or old system codes with no useful pattern, and one cost column simply repeats another. The project phase column also has no link to whether a project is delayed. We drop these and explain why.
Undefined objective	The brief does not state the exact goal; we have to define it ourselves. So we clearly write down the targets we chose, the columns we left out, and the assumptions we made.

2. Background Study / Literature Review

Cost overrun risk assessment[3] paper highlights the different models and its comparison with the training and evaluation models used. Multiple papers suggest neural networks [1][5]. Different pre-processing steps and feature engineering with different models are detailed in paper[6]. The references section contains the list of literature reviews attached. Different models used in research papers attached commonly include Random Forest, ANN, Logistic Regression, SVM and Ensemble[11].

3. Solution Approach

Our key business objectives include prediction of project delay, cost overrun and safety incident occurrence. We need to identify key drivers behind predicted risk for decision support.

Problem Statement	Model(s)	Tolerance to Bias	Tolerance to variance	Validation technique	Evaluation metrics
Predict Project Delay (Binary classification)	RF, NB, XGB, NN with softmax	Low - should avoid false negatives ie missing delayed projects	Medium - high Variance acceptable if early warnings are provided for high-risk projects	K-fold cross validation	F1, recall, ROC-AUC, cross entropy
Cost prediction	SVR, XGB, linear regression, ANN, Ensemble	Low to medium - as it involves cost predictions, needs to be more accurate	Low - finance forecasting	K-fold cross validation	R2, RMSE
Safety incident occurrence	RF, XGB, Ensemble, NN with softmax	Low - as it involves safety	Medium - false alarms can be acceptable	K-fold cross validation	Recall, F1, Cross-entropy

4. Workflow Diagram

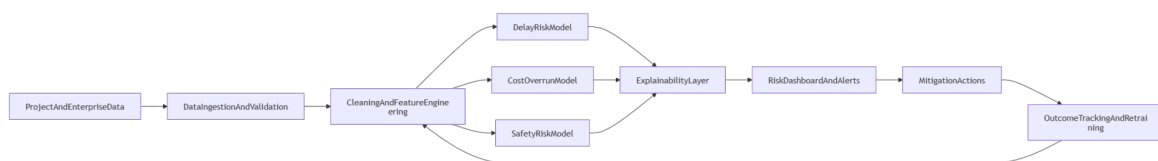


Fig 1.1 Model Workflow

The workflow transforms historical data into early-warning risk predictions with driver-level explanations, allowing project, procurement, finance, and safety teams to intervene before delays/cost escalations become severe.

5. Proposed Timeline

Activities	Timeline
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Pre-submission document	28th June, 2026
Domain analysis and data visualization	12th July, 2026
Feature engineering and Pre-processing completion	26th July, 2026
Machine learning modeling	30th Aug, 2026
Metrics comparison and hyper parameter tuning	30th Sep, 2026
Business recommendations and predictive analysis	30th Oct, 2026
Final presentation and documentation	30th Nov, 2026

6. Expected Deliverables / Outputs

As an outcome of our project, we will produce the following deliverables

- a. Technical documentation including the following
 - i. Data visualization and analysis
 - ii. Machine learning model(s) used
 - iii. Metrics comparison between the different machine learning models
 - iv. Dashboard for project-level and portfolio-level risk review
 - v. Business recommendations and mitigation
- b. Github repository link(s)
- c. Presentation with the summary of approach, results and recommendations

7. References

1. [Machine Learning Application in Construction Delay and Cost Overrun Risks Assessment](#)
2. [Machine learning applications for predicting safety incidents in construction industry](#)
3. [A Comparative Assessment of Concrete Compressive Strength Prediction at Industry Scale Using Embedding-Based Neural Networks, Transformers, and Traditional Machine Learning Approaches](#)
4. [Cost Overrun Risk Assessment and Prediction in Construction Projects: A Bayesian Network Classifier Approach](#)
5. [Leveraging AI for Predicting Delay Risks in Construction Project Management: Emerging Trends and Research Gaps](#)
6. [Predicting the duration of goods transportation delays based on machine learning methods](#)
7. [Predicting the Compressive Strength of High Strength Concrete Using Machine Learning Algorithms: A Comparative Analysis of Regression Models -](#)
8. [Deep Multilayer Perceptron Neural Network for the Prediction of Iranian Dam Project Delay Risks](#)
9. [Real-time prediction of project duration deviation using CNN-LSTM-FIS](#)
10. [Predicting the Indirect Cost of Construction Projects in Egypt: An Artificial Neural Network Approach](#)
11. [Ensemble learning framework for forecasting construction costs](#)
12. [Predicting cost overrun in construction projects using machine learning algorithms: the case of Jordan](#)